Data-Driven Color Economics: Decoding New Growth Signals in the Electric Vehicle Sector

Final Strategic Report

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Abstract

This research addresses the increasing commoditization within the Electric Vehicle (EV) industry by exploring "alternative data" beyond financial statements to assess a company's potential value and strategic maturity. I propose and validate an innovative analytical framework that leverages automotive color, a traditionally aesthetic element, as a quantifiable business strategy signal. The research process began with the quantification of thousands of unstructured color names by applying the K-Means clustering algorithm in the HSV color space, successfully creating a unified 15-dimensional color vector system. Building on this foundation, I constructed and optimized a multi-output XGBoost classification model, effectively addressing class imbalance through a custom data augmentation strategy. The final model demonstrated excellent performance, achieving a Jaccard Similarity Score of 0.7850, which confirms a highly predictable and strong correlation between vehicle parameters and color portfolio choices.

The core contribution of this report lies in the deep interpretation of the model, which reveals a three-tiered logic of "Brand-Positioning-Attributes" influencing color decisions. Furthermore, by combining SHAP value analysis with real-world market cases (e.g., BYD, Tesla, Wuling), I have successfully decoded the specific business strategies behind colors such as red (performance tag), blue (value-tech), and purple (brand exploration). Ultimately, this research provides not only a viable "Color Economics" analytical method but also a concrete, actionable decision-support framework for due diligence and market opportunity identification in the venture capital domain.

1. Introduction

The Electric Vehicle (EV) industry has entered its second phase of competition. As technology converges and markets saturate, traditional competitive advantages are diminishing. For forward-thinking investors

like Qualcomm Ventures, the challenge is to penetrate beyond the fog of financial statements to identify non-traditional signals that can predict a company's future growth potential. This report addresses that challenge by transforming automotive color—a traditionally subjective aesthetic element—into a quantifiable, interpretable strategic assessment tool. I contend that a company's color strategy is a direct reflection of its core competencies in brand positioning, market understanding, and cost control. This report will detail the data science framework I built to decode this "color language" and, ultimately, provide a decision-making compass for investors to gain a unique advantage in due diligence and market opportunity identification.

2. Methodology and Process

The success of this research is built upon a systematic, rigorous, and iteratively optimized data science workflow that fully covers the process from raw data to final business insight.

2.1 Foundational Work: The Quantification of Color Systems

The project's initial core challenge stemmed from the unstructured nature of the data: thousands of descriptive color names created by manufacturers (e.g., "Stardust Blue," "Glacier White"). To resolve this, I first needed to establish a unified, quantitative standard for "color." Our methodology included two key steps: first, converting all RGB color values to the HSV (Hue, Saturation, Value) color space, which more accurately simulates human perception of color similarity by separating hue from saturation and brightness. Second, I applied the classic K-Means clustering algorithm in this space to objectively and reproducibly converge thousands of discrete color data points into 15 representative core color clusters. This pioneering work, completed in the project's midterm, successfully transformed chaotic text descriptions into a uniform, structured 15-dimensional color vector, laying the data foundation for all subsequent modeling.

2.2 Model Construction and Optimization

With a structured data foundation, I proceeded to the predictive modeling phase. To align the model more closely with the business decision of "offering/not offering," I transformed the target variables from color option counts to a binary "0/1" format, precisely defining the problem as a multi-label classification task.

Facing the common issue of severe class imbalance in real-world data, I designed and implemented a custom data augmentation strategy based on normal distribution. This strategy learned and resampled the feature distributions of minority classes, creating diverse synthetic data while preserving their essential characteristics, effectively balancing the training set and preventing model bias.

Finally, I selected the XGBoost algorithm, renowned for its excellence in handling tabular data, as the

core model. It was wrapped in a MultiOutputClassifier to train a dedicated classifier for each of the 15 color clusters. This architecture ensured that the model could handle the complex multi-target prediction task while maintaining a powerful ability to capture non-linear relationships and accommodate the uniqueness of each specific color.

3. Results & In-depth Analysis

3.1 Model Performance: High-Confidence Predictive Capability

The optimized final model demonstrated excellent performance. **According to the evaluation results**, the model achieved a **Jaccard Similarity Score of 0.7850**. This metric measures the overlap between the model's predicted "color portfolio" and the actual one, with a score of 0.7850 indicating an average similarity of 78.5%. Concurrently, a **Hamming Loss as low as 0.0646** signifies an average error rate of just 6.5% across all individual "available/unavailable" judgments. These complementary metrics together prove that EV color selection is not a random aesthetic act but a highly predictable business decision correlated with vehicle parameters and market strategy, providing a solid statistical foundation for the reliability of all subsequent analyses in this report.

3.2 Global Insights: The Three-Tiered Logic of Color Decisions

To systematically understand the universal laws governing color selection, I conducted a "peeling-the-onion" style analysis of feature importance.

First, **in the Figure 3-1**, I observed that in the initial analysis including all features, those prefixed with Manufacturer_ (e.g., Tesla, BYD) dominated the importance rankings. This leads directly to our first conclusion: **Brand is the supreme commander of color**. Color is a direct externalization of brand DNA, reflecting its market positioning, target audience, and even production philosophy.

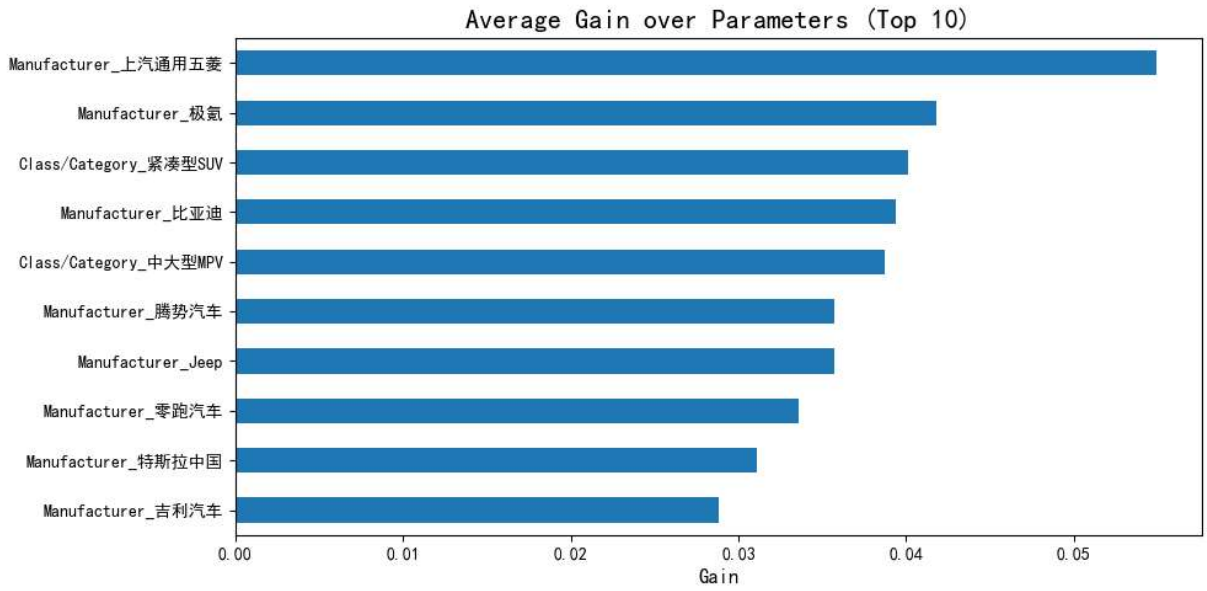

Figure 3-1 Average Gain over Features

Second, **in the Figure 3-2**, when I stripped away the brand effect, features prefixed with Class/Category_ (e.g., Mid-size SUV, Compact Car) rose to prominence. This data phenomenon indicates that within the same brand, color strategies differ for product lines targeting different market segments. Therefore, our second conclusion is: **Market segmentation is the battlefield for color**.

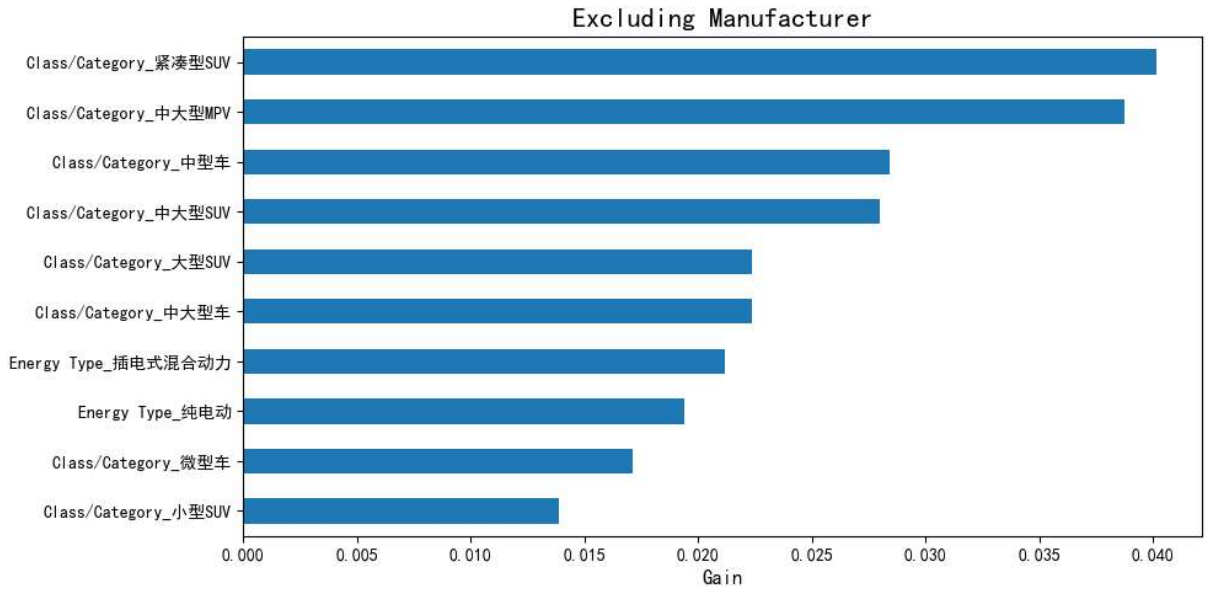

Figure 3-2 Average Gain over Features (Excluding Manufacturer)

Finally, **in the Figure 3-3**, when I further excluded brand and vehicle class, I saw the ultimate constraints on color choice. Hard parameters like suggested retail price, curb weight, and max power became the most important drivers. From this, I draw our third conclusion: **Physical and economic attributes are the bedrock of reality**. Color ultimately serves the constraints of cost control, performance expression, and technological positioning.

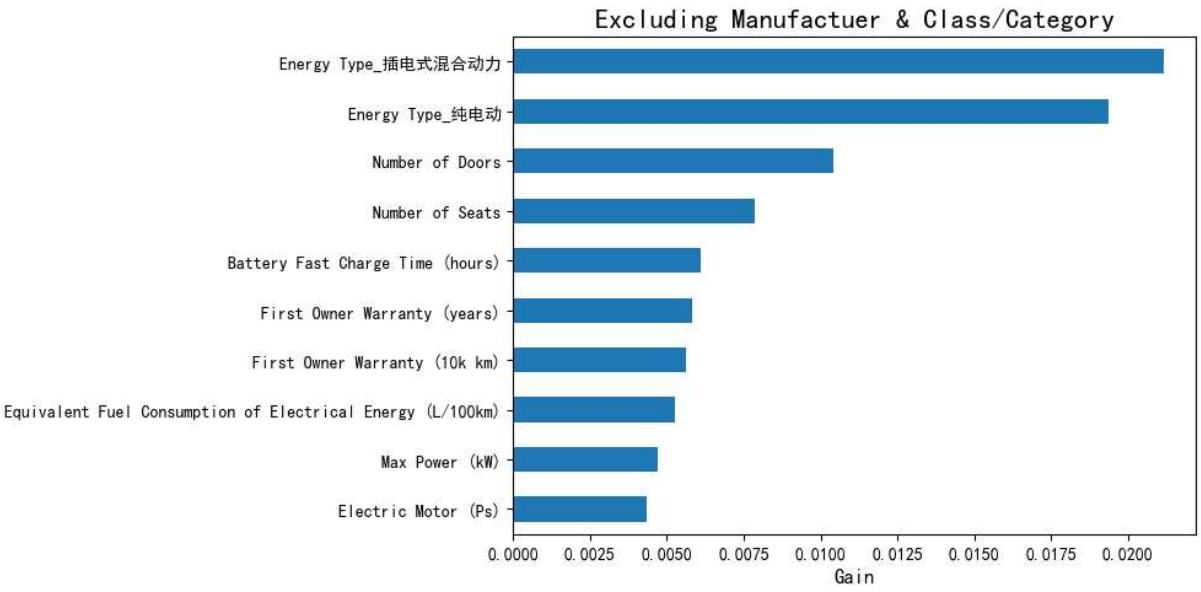

Figure 3-3 Average Gain over Features (Excluding Manufacturer&Class/Category)

3.3 Profiling Colors: Five Real-World Business Strategies

Through SHAP (SHapley Additive exPlanations) value analysis, I were able to "profile" specific colors and reveal their distinct business strategies.

Red: The Classic Performance Tag Strategy

Data Description: In the SHAP plot from Figure 3-4, Max Power (kW), Electric Motor (Ps), and Max Torque (N·m)—three metrics that directly reflect performance—rank at the top of all features. The plot clearly shows that the higher the values of these features (dots colored red), the higher their corresponding SHAP values (dots shifted to the right), providing a strong positive push for the offering of the color red.

Inference & Analysis: The data indicates that the higher a vehicle's performance, the more likely it is to be offered in red. In the EV era, red continues its legacy as a visual symbol of high performance, sportiness, and driving passion. It is not a mass-market color but a precise tag used by automakers to identify their high-performance product lines and attract driving enthusiasts, a strategy in line with a century of automotive cultural heritage.

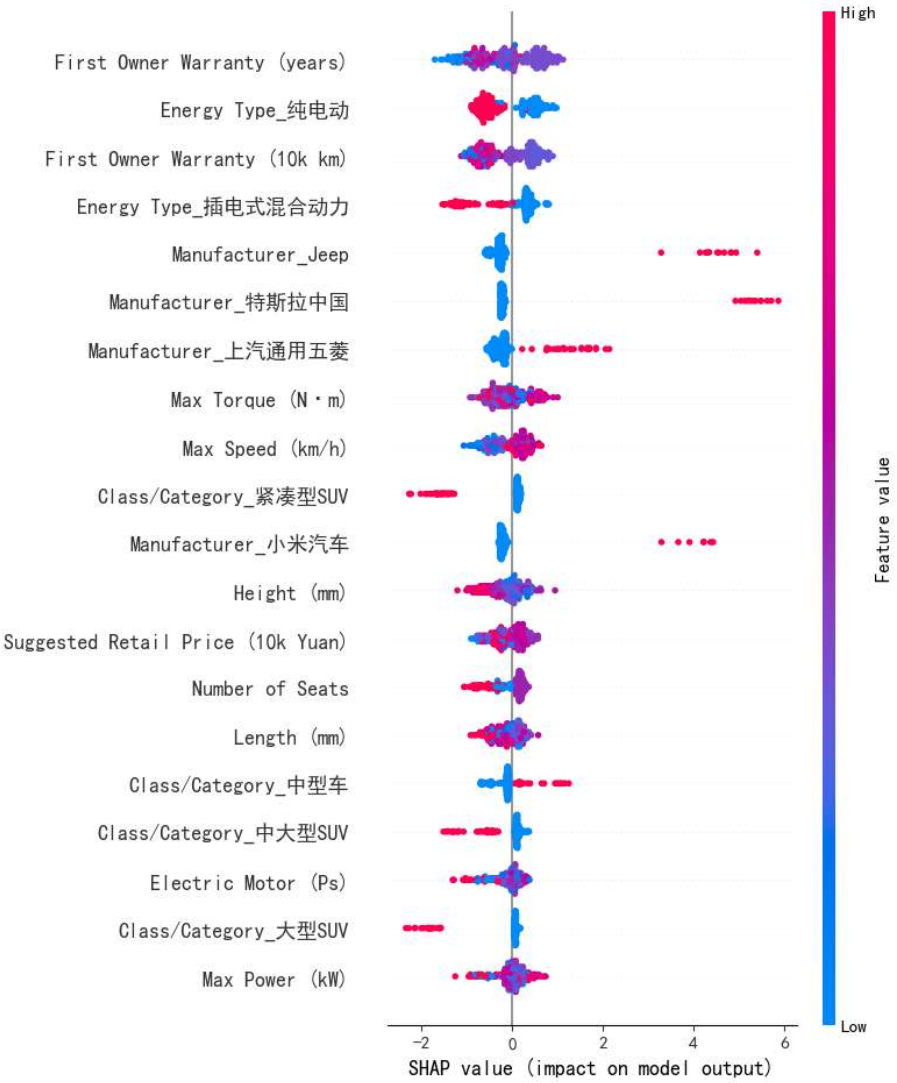

Figure 3-4 SHAP for Red Color

Ocean Blue: The Value-Technology Strategy

Data Description: The SHAP plot in Figure 3-5 shows that Suggested Retail Price and CLTC Pure Electric Range are two key drivers. The plot indicates that the lower the price (dots colored blue) and the higher the range (dots colored red), the higher the corresponding SHAP value, increasing the probability of Ocean Blue being offered.

Inference & Analysis: The data shows that Ocean Blue is more often found on vehicles that are affordable yet have an outstanding range. This aligns perfectly with the successful strategy of BYD's "Ocean Series." Ocean Blue is a typical "value-tech" signal color, used by BYD to convey a core value proposition to the mainstream market: "I'm affordable, I go far, and I'm cool."

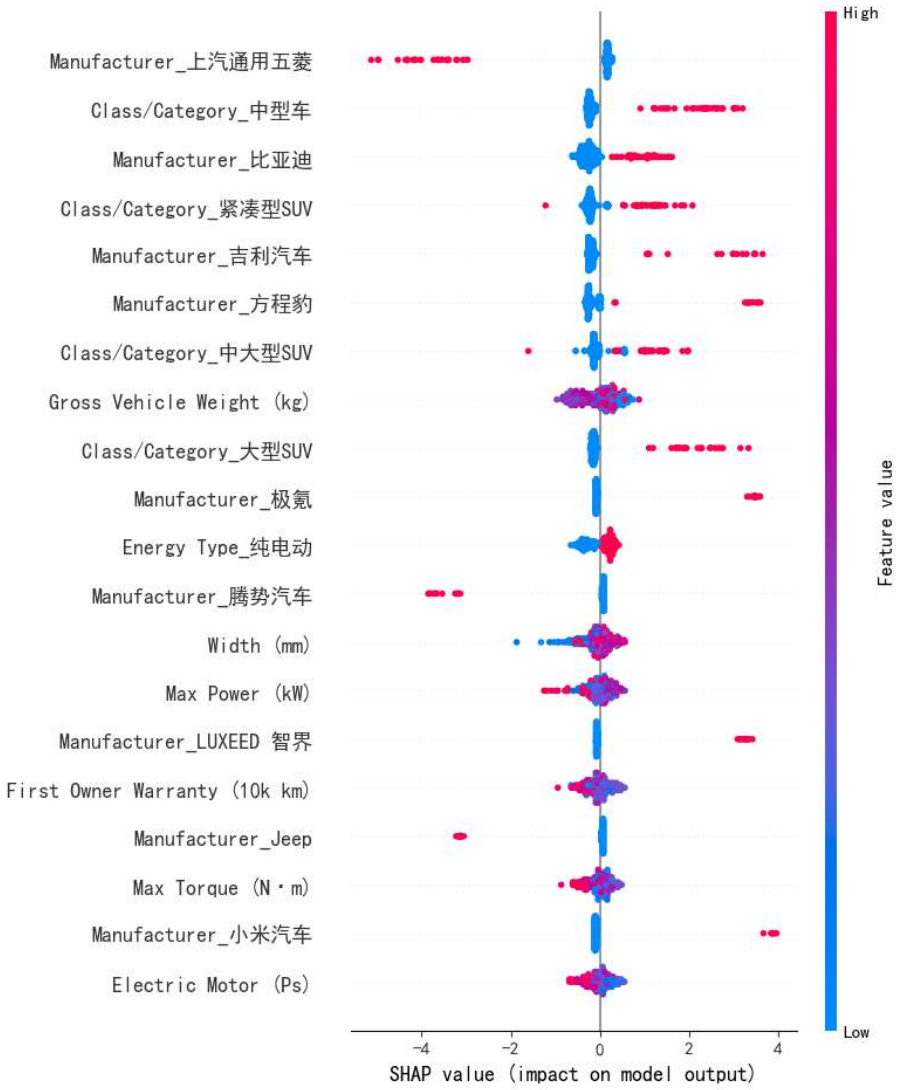

Figure 3-5 SHAP for Ocean Blue Color

Green & Purple: Tools for Differentiation

Data Description: The SHAP plot for Green in Figure 3-6.png shows it is more associated with smaller vehicle sizes and weaker acceleration. In contrast, the SHAP plot for Purple in Figure 3-7 shows its drivers are more scattered, linked to specific models from various brands.

Inference & Analysis: These two colors represent automakers' exploration of differentiation beyond the main palette. Green represents concepts of "economy" and "eco-friendliness." Purple, on the other hand, is not a universal market color but a flexible tool used by brands for niche exploration, such as creating limited or special editions.

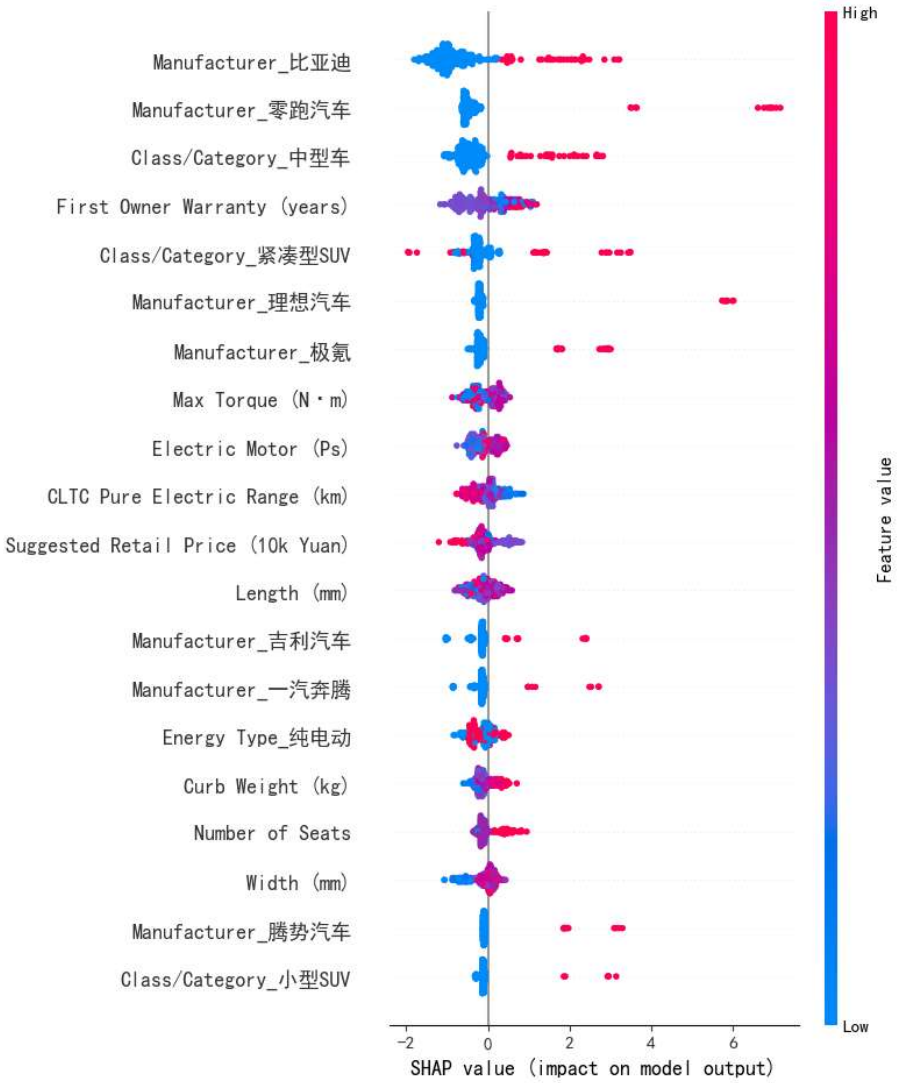

Figure 3-6 SHAP for Green Color

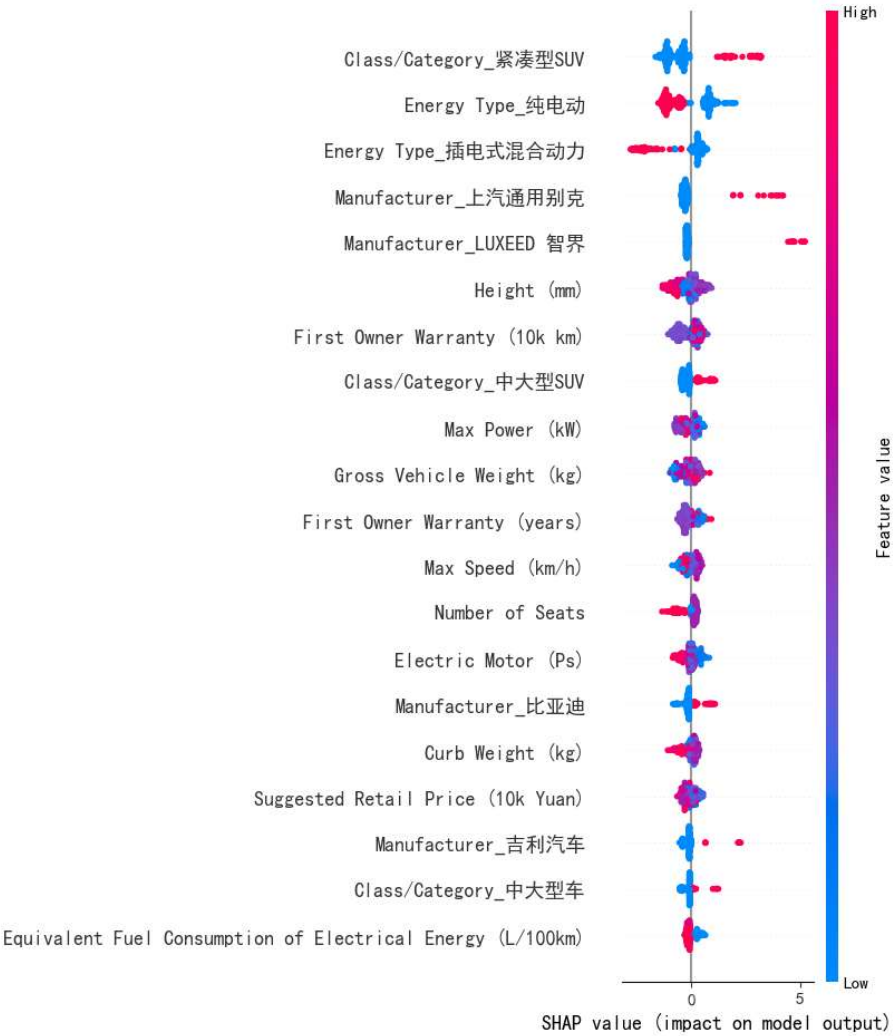

Figure 3-7 SHAP for Purple Color

Grey: The Mass-Market Bedrock and a Statistical Trap

Data Description: The SHAP plot for Grey in Figure 3-8 shows a peculiar pattern where nearly all features appear to positively drive the offering of Grey.

Inference & Analysis: At first glance, this might suggest that Grey is a universally "versatile" color. However, cross-validating this with model evaluation data from output.txt, I find that in the test set, there are only 8 samples that do not offer Grey. This extreme imbalance led the model to adopt a speculative "always predict Grey" strategy. Thus, the "universality" shown in the SHAP plot is a statistical artifact caused by data distribution. The true conclusion is: Grey is the absolute bedrock & default color of the current market, so prevalent that not offering it has become a rare decision that requires explanation.

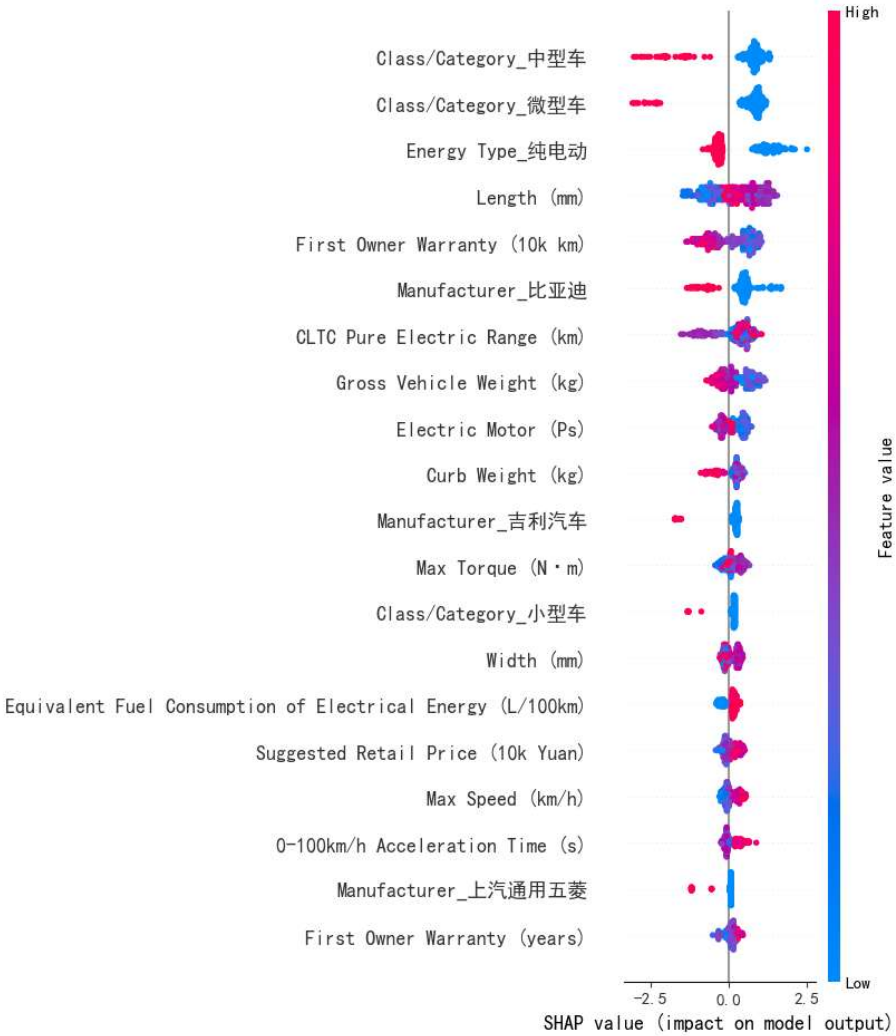

Figure 3-8 SHAP for Grey Color

4. Strategic Recommendations for Venture Capital

Based on the analysis supported by data and real-world business cases, I provide a concrete decision-making framework for Qualcomm Ventures.

Due Diligence via "Strategic Coherence" Test

A mature company's product decisions should be highly aligned with its strategic positioning. Our data

analysis provides a unique tool to verify this. For example, our data shows that "Green" is generally associated with economy, small-sized, and lower-performance vehicles. However, **XPeng** markets a "Stardust Green" for its high-performance P7i model, attempting to link the color to "intelligent performance." This does not mean XPeng is wrong, but it constitutes a risk point that must be deeply questioned during due diligence. The team can ask: "Does your company have sufficient brand power and marketing budget to reshape the market's traditional perception of Green?" In this way, color analysis is transformed from an observation into a potent due diligence tool.

Market Opportunity Identification via "Strategy-Mix" Analysis

Our analysis can also reveal market gaps. The data shows that there are established strategy mixes like "Performance-Red" and "Economy-Commuter-Macaron." However, the broad middle market in between has a relatively conservative color strategy. This points to an overlooked opportunity: **Is there an automotive aftermarket startup (e.g., color wraps or custom painting) whose business model is to provide cost-effective, yet highly fashionable and personalized color upgrade solutions specifically for "value-tech" vehicles like those from BYD and GAC Aion?** This is a massive, data-indicated opportunity for scaling in the existing market.

Judging Ambition: "Market Follower" vs. "Market Definer"

Finally, color strategy can reveal a company's ambition. BYD's color strategy is a typical "market follower and innovator." Wuling, in contrast, successfully "defined" a new market segment through color. When evaluating a startup, its color strategy can help judge its potential: Is it hoping to take a slice of an existing market (a follower), or does it have the capability and plan to create a new category (a definer)? The valuation models and investment theses for these two types of companies are fundamentally different.

5. Conclusion

This research transformed automotive color from a subjective aesthetic issue into a quantifiable and interpretable business strategy. I not only validated the model's effectiveness but more importantly, through the combination of data and real-world business cases, I have revealed proven, successful color strategies. Ultimately, this project provides a practical, alternative-data-based decision-making tool for venture capital, enabling a deeper and more forward-looking assessment of investment opportunities and risks in the dynamic EV sector.